Sultanate of Oman
Ministry of Health
The Royal Hospital
Department of Child Health

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Title: Riboflavin Supplementation

Riboflavin is well known chaperone effective in the treatment of inborn error of metabolism

Indication of treatment:

- **Riboflavin transporter deficiency neuronopathy (Brown-Vialetto-Van Laere (BVVL) syndrome.** This is an autosomal recessive condition due to pathogenic variants in either *SLC52A2* or *SLC52A3*. It is characterized by motor neuronopathy manifest as proximal and distal limb weakness, often with severe distal wasting and breathing problems due to paralysis of the diaphragm, sensory neuronopathy (manifest as gait ataxia), and cranial neuronopathy (manifest as optic atrophy, sensorineural deafness, and bulbar palsy.
- **A riboflavin-responsive lipid storage myopathy due to multiple acyl-CoA dehydrogenase deficiency**: autosomal recessive condition manifested as muscle weakness and myopathy in adult population
- **Riboflavin responsive dihydrolipoamide dehydrogenase deficiency: This condition manifesting as manifesting progressive exertional fatigue associated with metabolic acidosis and hyperlactemia**

Treatment protocol

Because oral riboflavin supplementation is effective and possibly life saving, it should begin as soon as a riboflavin transporter deficiency neuronopathy is suspected and continued lifelong unless the diagnosis is excluded by genetic testing

High-dose supplementation of riboflavin between 10 mg and 50 mg/kg/day has been effective in patients with genetically confirmed riboflavin transporter deficiency neuronopathy and also in some patients in whom the genetic basis of riboflavin transporter deficiency neuronopathy has not been established. It is given in three divided doses.

Note: Some patients are on doses up to 70 mg/kg/day.

For the riboflavin responsive multiple acyl-COA dehydrogenase deficiency and riboflavin responsive dihydrolipoamide dehydrogenase deficiency dose of 10/g/kg/day is recommended.

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